



Algorithm Analysis & Design

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How to ace the course?



12 Weeks Course (excluding exam/holiday/prep weeks)

Total Time per week: 10 hrs

Session	Time (hrs)
2 Lectures	2 x 1.5
1 Tutorial	1 x 1
Reading/Practice	3
Problem Solving	3

Evaluation



Evaluation	Marks
Quiz x 2	10 x 2
Mid Sem	25
End Sem	45
Assignments (Peer Evaluated)	10

References



Algorithms by S. Dasgupta, C. H. Papadimitriou, and U. V. Vazirani

<http://algorithmics.lsi.upc.edu/docs/Dasgupta-Papadimitriou-Vazirani.pdf>

Algorithms by Jeff Erickson

<https://jeffe.cs.illinois.edu/teaching/algorithms/#notes>

Algorithms Illuminated by Tim Roughgarden

<https://github.com/ailyanlu1/algorithm-specialization-notes/blob/master/Algorithms%20Illuminated%20Part%201%20The%20Basics.pdf>

Algorithm Design by Kleinberg & Tardos

<https://theswissbay.ch/pdf/Gentoomen%20Library/Algorithms/Algorithm%20Design%20-%20John%20Kleinberg%20-%20C3%89va%20Tardos.pdf>

Design & Analysis of Algorithms by Dexter Kozen

<https://www.cs.cornell.edu/~kozen/Papers/daa.pdf>



Numbers & Algorithms

Lecture 1 | Algorithms Design & Analysis

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Numbers



- Humans evolved from hunter-gatherers to farming communities
- Barter Trade started to happen.
- The need for working with large numbers arose



Expressing Large Numbers



- Number alphabets are developed from 0 to base - 1
- Larger numbers are expressed as a string of digits
- Several bases were popular
 - Base 60 (Sumerians, Babylonians)
 - Base 10 (Hindu-Arabic)



Adding Numbers



$$\begin{array}{r} \text{Carry: } 1 \qquad \qquad \qquad 1 \quad 1 \quad 1 \\ \qquad \qquad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad (53) \\ \qquad \qquad 1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad (35) \\ \hline 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad (88) \end{array}$$

Multiplying Numbers



input consists of a pair of arrays $X[0..m-1]$ and $Y[0..n-1]$, representing the numbers

$$x = \sum_{i=0}^{m-1} X[i] \cdot 10^i \quad \text{and} \quad y = \sum_{j=0}^{n-1} Y[j] \cdot 10^j,$$

and similarly, the output consists of a single array $Z[0..m+n-1]$, representing the product

$$z = x \cdot y = \sum_{k=0}^{m+n-1} Z[k] \cdot 10^k.$$

Multiplying by Repeated Addition



Correctness: By definition

Runtime: $Y \times m = 2^n \times m$

Correctness & Runtime Analysis



Correctness: Algorithm computes bellow formula, with holds to ensure that the output is a valid digit string with a 10 ary expansion

$$x \cdot y = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} (X[i] \cdot Y[j] \cdot 10^{i+j}).$$

Runtime: $n \times m$

Read Section 1.1.2 in [1]

Solve Exercise 1.8 in [1]

[1]

<http://algorithmics.lsi.upc.edu/docs/Dasqupta-Papadimitriou-Vazirani.pdf>

Asymptotics



See first half of [1]

See Section 0.3 in [2]

Solve Exercise 0.1 in [2]

[1] <https://users.cs.duke.edu/~reif/courses/alglectures/demleis.lectures/lec2.pdf>

[2] <http://algorithmics.lsi.upc.edu/docs/Dasgupta-Papadimitriou-Vazirani.pdf>

Euclid's GCD



$\text{GCD}(a, b) = \text{GCD}(b, r)$ Why?

$$a = b q + r$$

Any common divisor of a , b should divide r .

See Section 1.2.3 in

<http://algorithmics.lsi.upc.edu/docs/Dasgupta-Papadimitriou-Vazirani.pdf>

Pingala-Fibonacci Numbers



See Section 0.2 in

<http://algorithmics.isi.upc.edu/docs/Dasgupta-Papadimitriou-Vazirani.pdf>

Reading



Section 0.2, 0.3, 1.1, 1.2.3 in

<http://algorithmics.lsi.upc.edu/docs/Dasgupta-Papadimitriou-Vazirani.pdf>

Chapter 0 in

<https://jeffe.cs.illinois.edu/teaching/algorithms/book/00-intro.pdf>

Algo Design Idea I: Recursion



MergeSort



Algo Design Idea II: Divide & Conquer



Running Time



Master Theorem for Recursion



Karatsuba's Multiplication Algorithm

